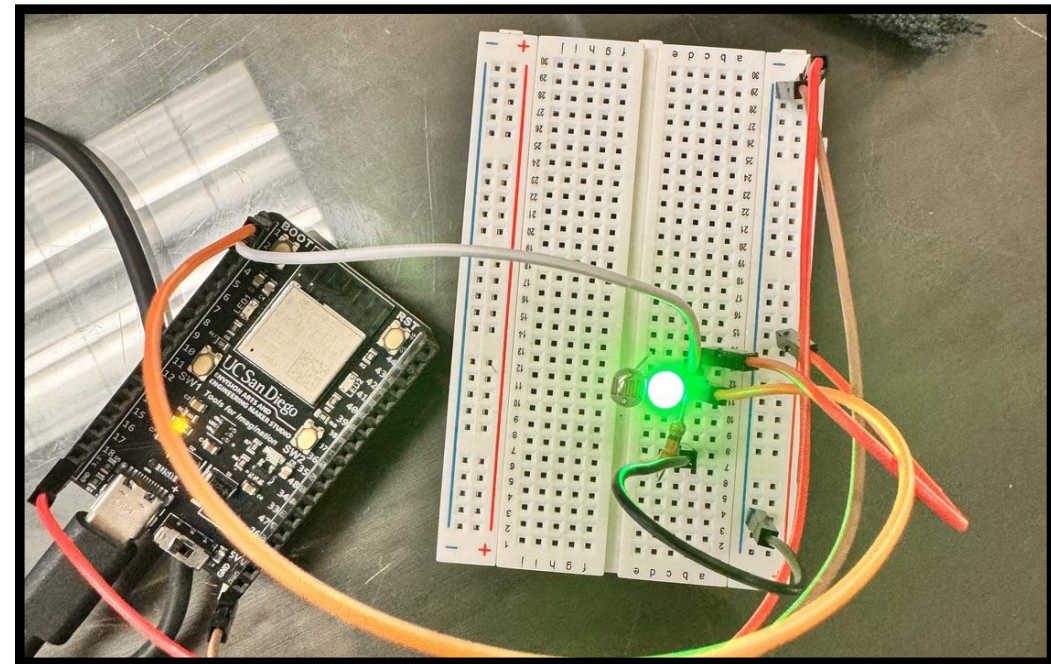


Lab 0: Microcontrollers



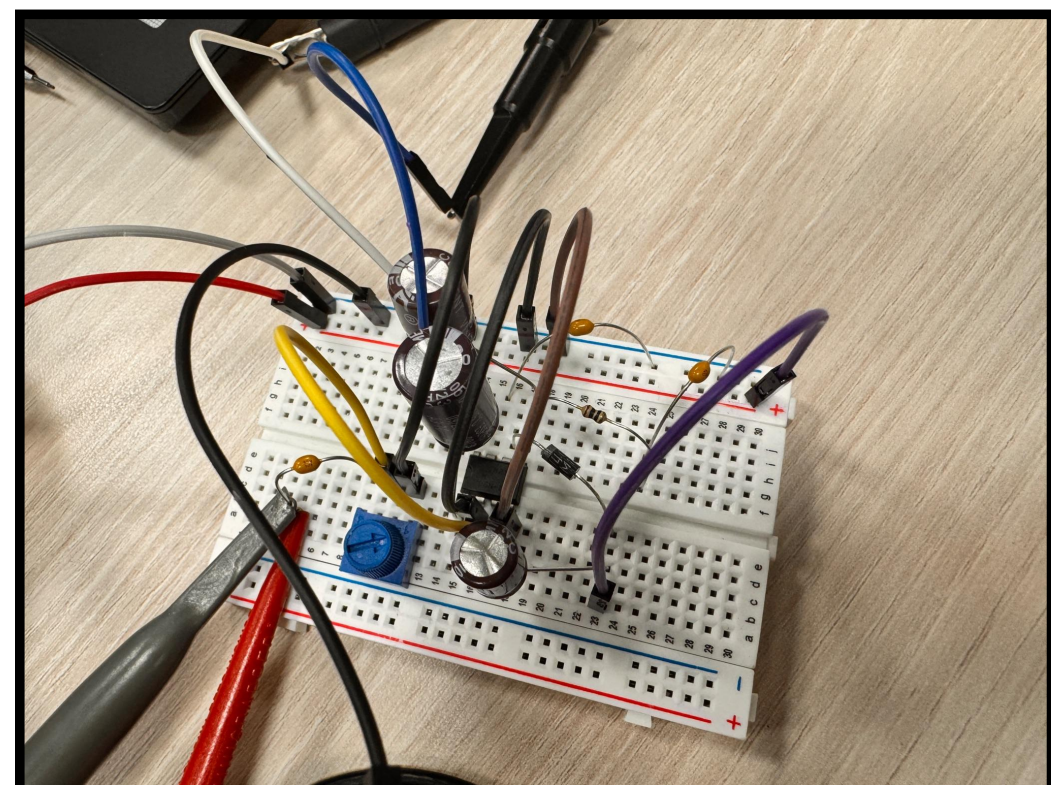
This lab introduced the ESP32 microcontroller and Arduino IDE. We learned to program the ESP32 to control LEDs, read buttons, and use a photoresistor via serial communication.

Lab 1: Communication



We used the ESP32 to build a wireless web server over Wifi. The ESP32 acted as an access point. We then connected our client device to it through HTTP to display options on a webpage that would allow us to manipulate an RGB.

Lab 2: Analog Circuits



We built and tested analog circuits using Op-Amps and created an amplifier, low-pass filter, high-pass filter, and band-pass filter. We also then analyzed signals in both the time and frequency domains with an oscilloscope.

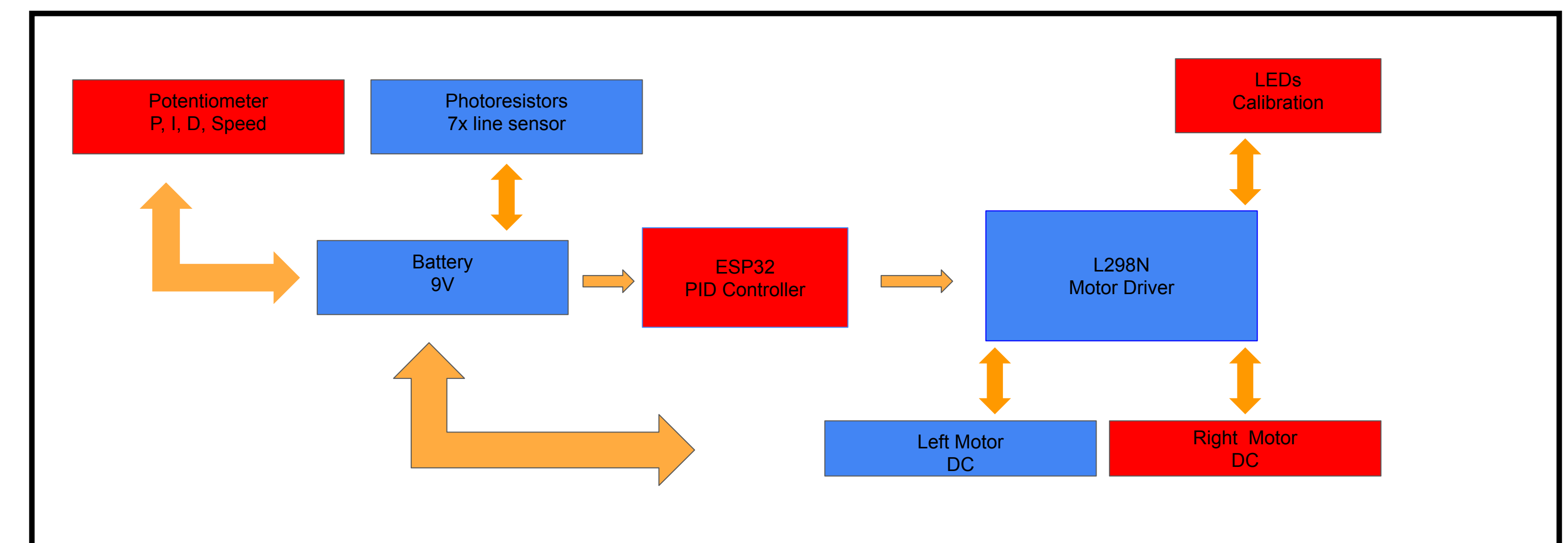
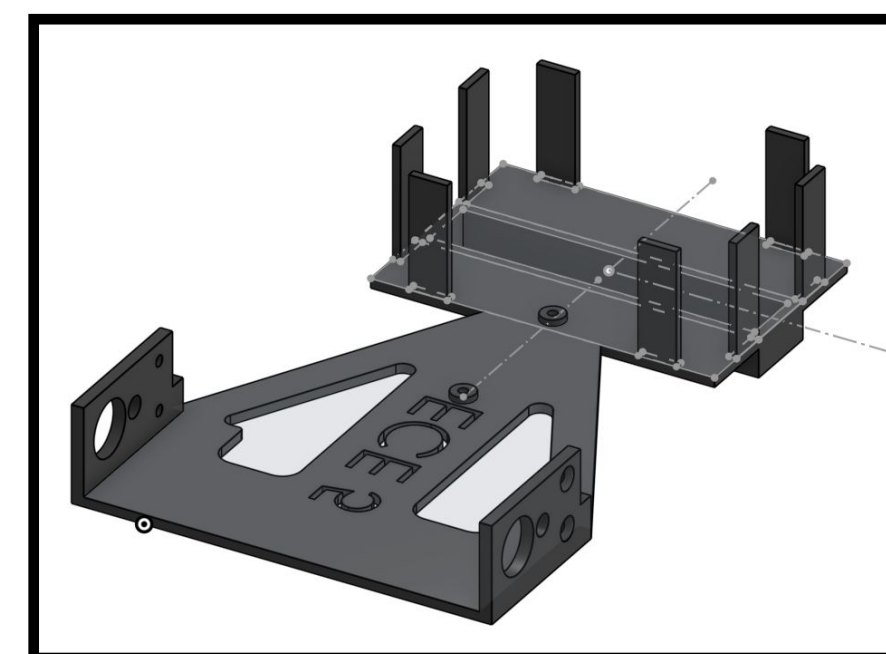
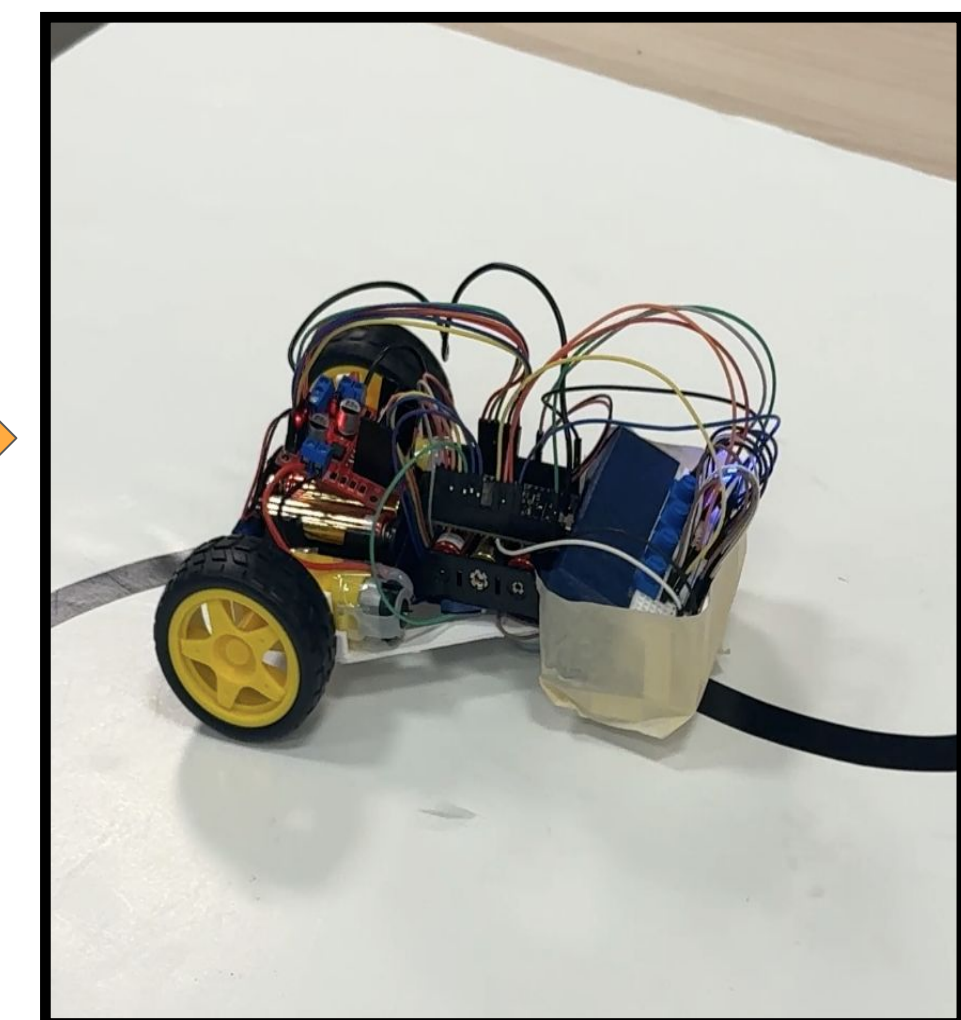
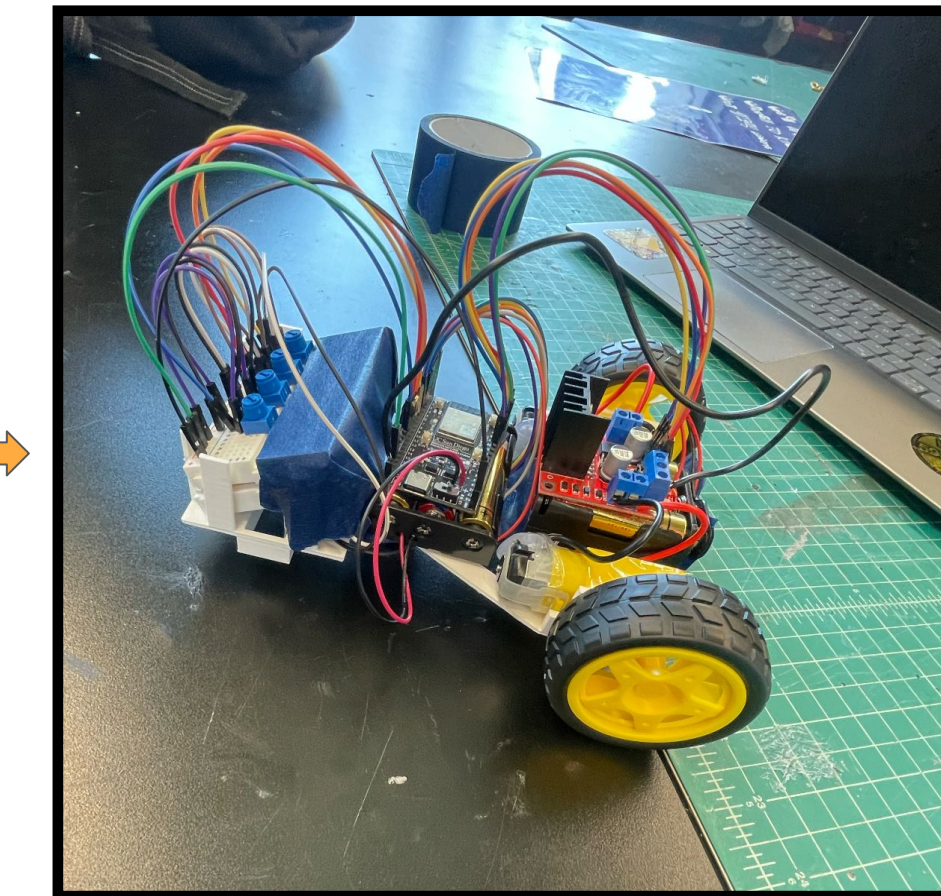
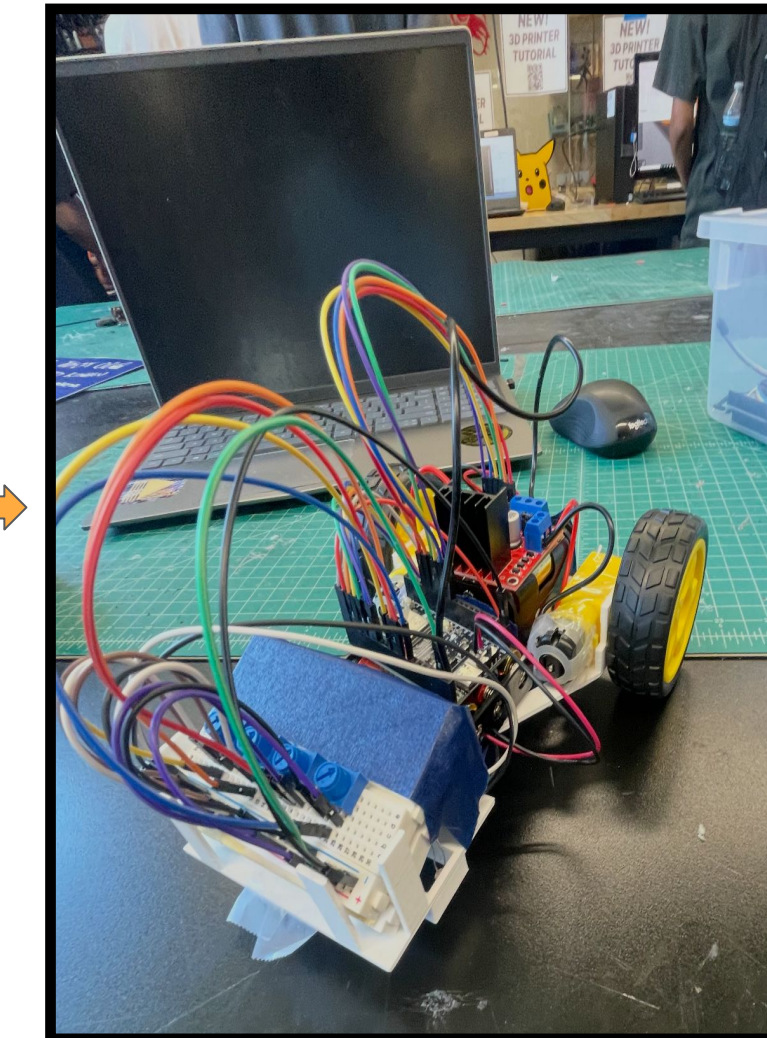
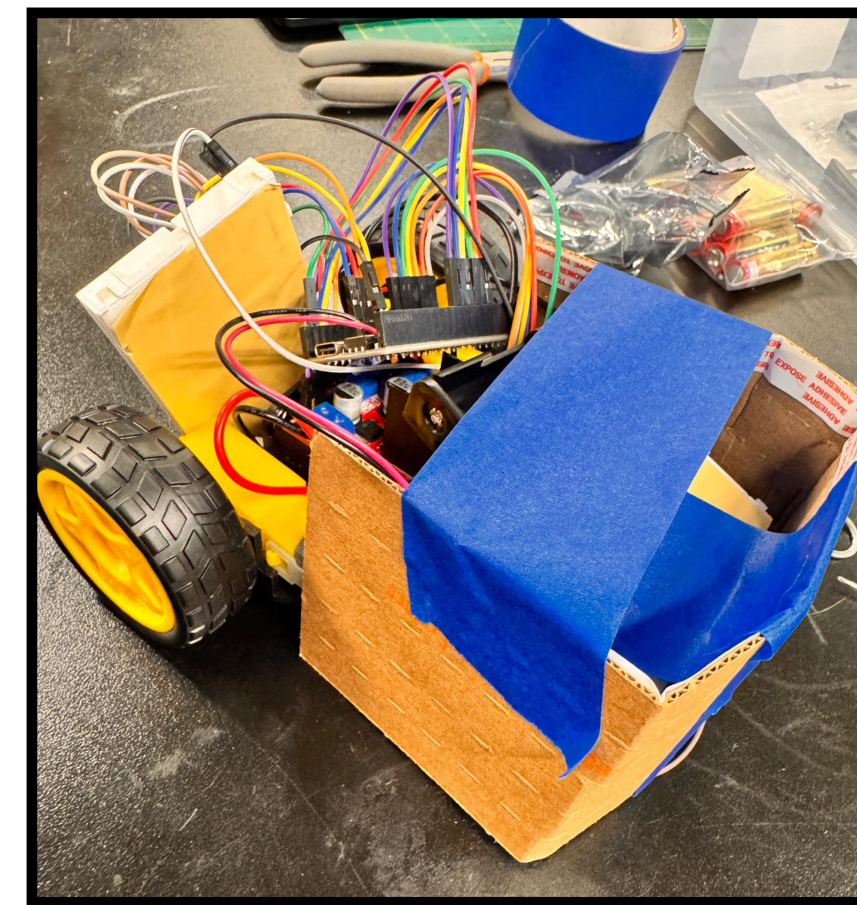
Lab 3: Digital Signal Processing



We used MATLAB to explore DSP concepts, including ADC, sampling, Fourier transforms, and spectrograms. We also then captured real-world photoresistor data through the ESP32 and processed and visualized it at different sampling rates.

Each lab built up to our final robot piece by piece. Lab 0 taught us to program the ESP32 which we later used the robot's brain. Lab 1 allowed us to learn the very basics of HTML and serial communications. Lab 2 gave us the analog circuit skills needed to wire the photoresistors. Lab 3 helped us to understand how to interpret sensor data. All this allowed us to build toward a robot capable of following a line.

Robot Progress from Prototype to Final Design



Week 1: We setup the potentiometers, photoresistors, and motor drivers individually.

Week 2: Worked on a CAD design and 3D printing the chassis but the Envision Makerspace was really busy which set the print back and our early prototype did not have the correct chassis.

Week 3: Fully assembled the robot and began integration testing. Had to replace a defective motor, motor driver, and broken ESP32 board. However, our robot was capable of following a line despite us still actively tuning our PID values as we prepare for competition.